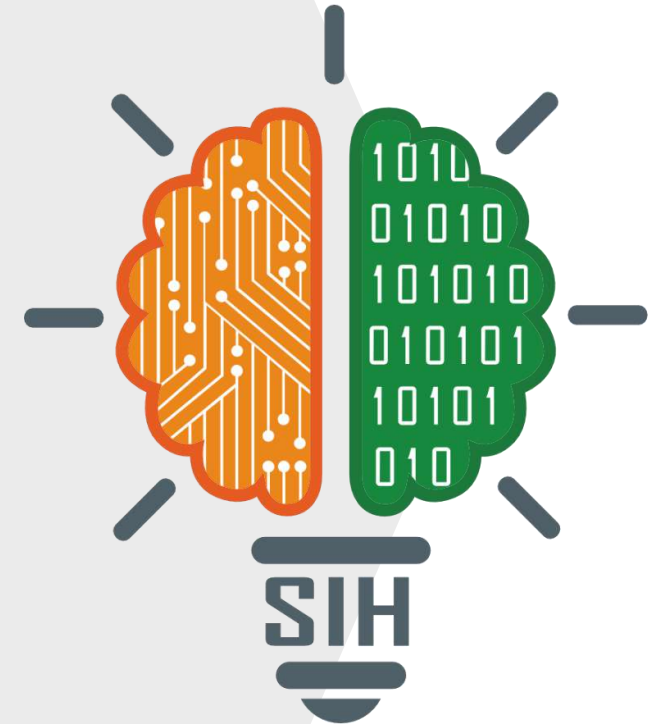


SMART INDIA HACKATHON 2025



ERP

- **Problem Statement ID – SIH25103**
- **Problem Statement Title-** ERP-based Integrated Student Management System
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** 51073
- **Team Name-** TLE_Smashers

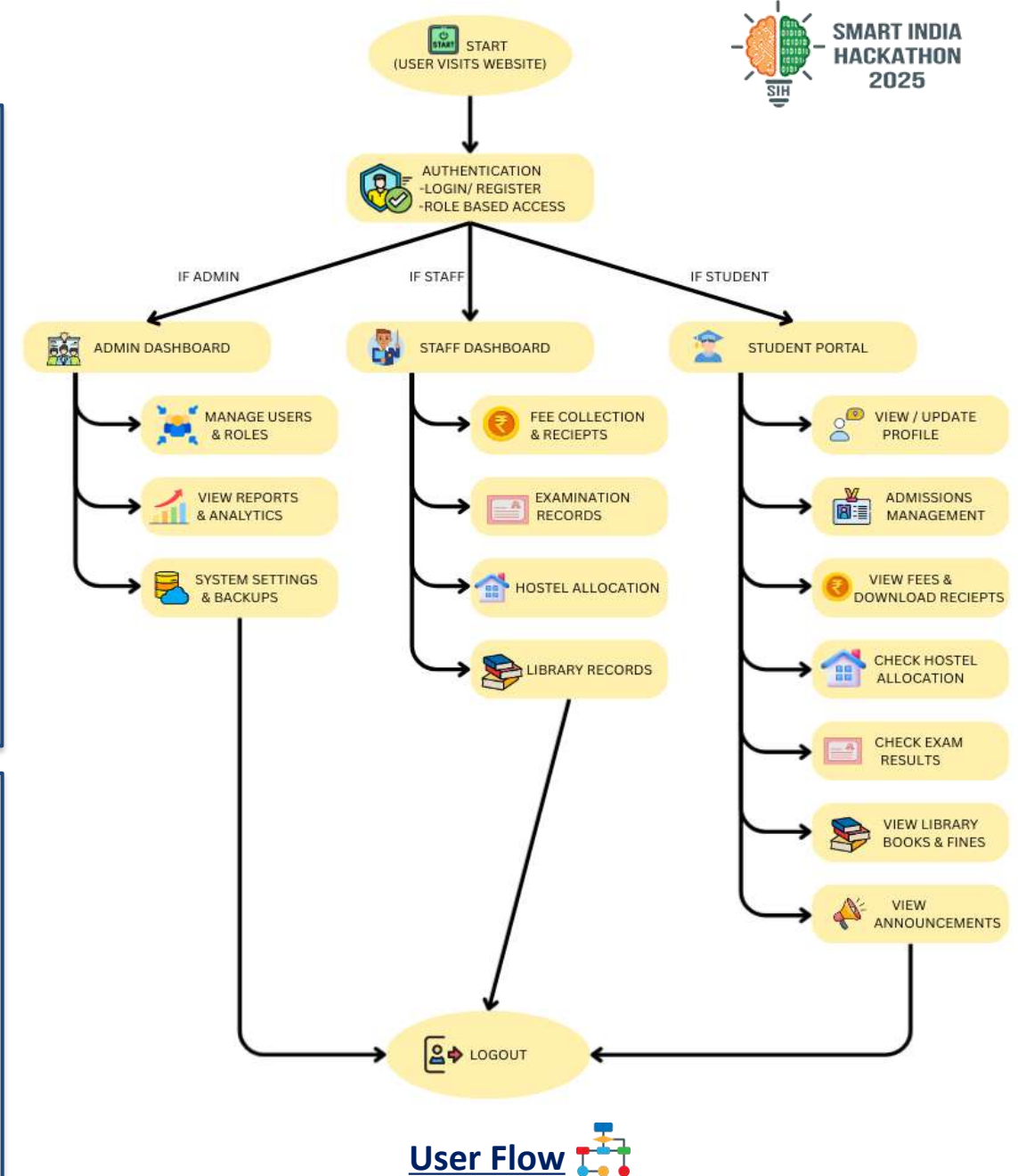


1 Proposed Solution (Prototype Live Link)

- One unified digital platform for admissions, fee collection, hostel, library, and exams.
- Replaces fragmented ledgers and manual queues with centralized, automated workflows.
- Single Student Database: One-time data entry, shared across departments.
- Automated Receipts & Allocations: Fees, hostel, library updates in real-time.
- Role-Based Access: Students, faculty, finance, hostel, librarian, exam cell, admin.
- Dashboards: Instant insights: admissions, revenue, hostel occupancy, exam performance.

Innovation & Uniqueness

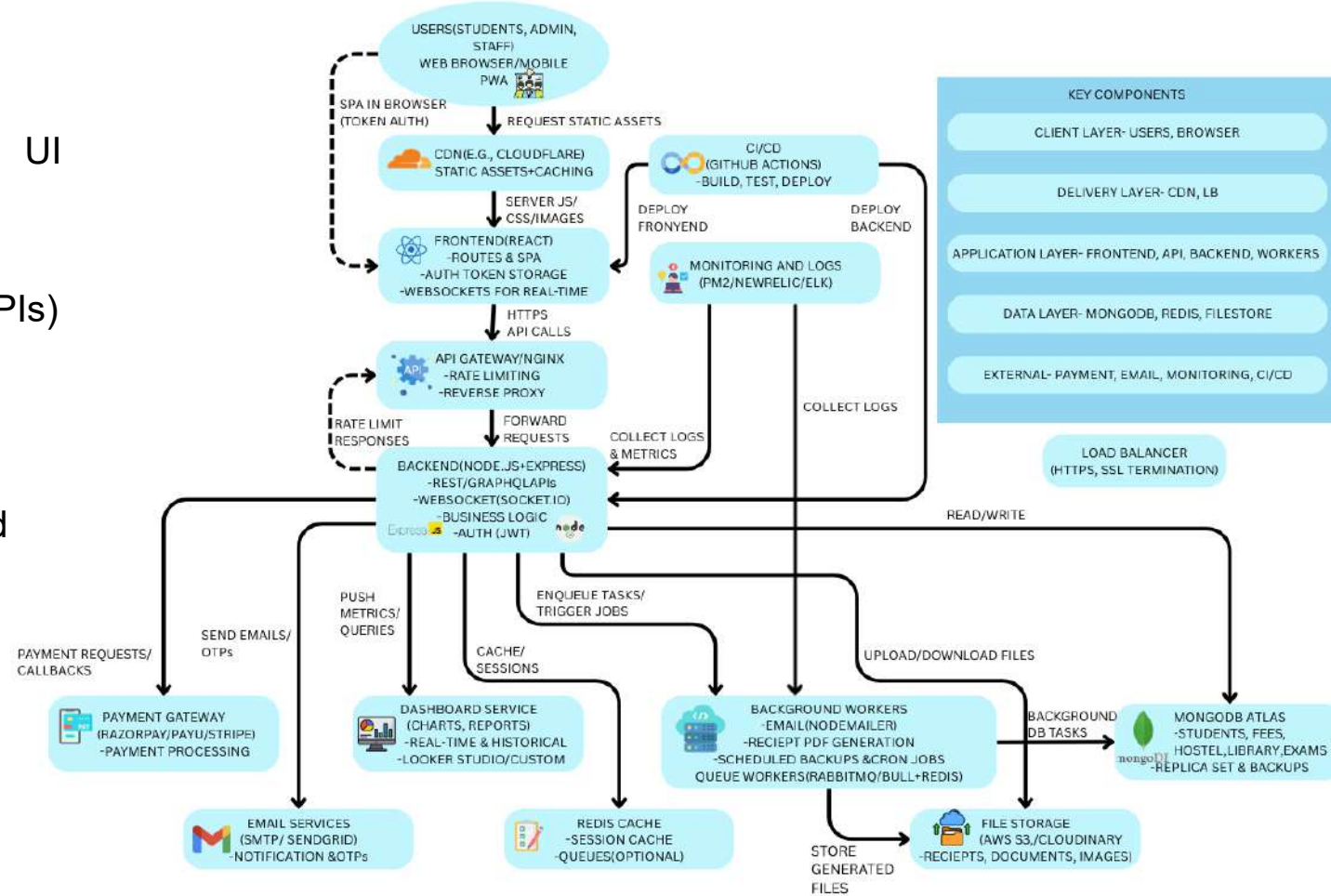
- **Affordability:** Built using open-source frameworks + familiar tools (forms, spreadsheets, dashboards).
- **Scalability:** Modular design; new features can be plugged in easily.
- **Ease of Adoption:** Minimal training needed, staff already familiar with spreadsheets & forms.
- **Replicable Model:** Any public college can adopt without huge investment.



2

Tech Stack

- **Frontend (Web):** React + TailwindCSS / Material UI (modern, responsive UI)
- **Backend:** Node.js + Express.js (lightweight, scalable APIs)
- **Database:** MongoDB Atlas (cloud-hosted)
- **Authentication:** JWT + bcrypt (secure login, role-based access)
- **Payments:** Razorpay / PayU / UPI integration (instant receipts, auto-updated records)
- **Deployment:** Docker, AWS, GitHub, GCP Vercel



3 Feasibility:

- Technical Feasibility

- ♦ Lightweight, cloud-based system → runs on low-cost servers
- ♦ Modular build (Admissions, Fees, Dashboard → extendable to Hostel/Library/Exams)
- ♦ APIs for easy integration with future/legacy systems

- Economic Feasibility

- ♦ Low-cost alternative to heavy ERP systems
- ♦ Affordable for public colleges (no expensive licenses)
- ♦ Can be maintained by in-house IT team or student interns

- Social Feasibility

- ♦ Simplifies processes for students (no queues, online access)
- ♦ Reduces staff workload & duplication

4 Viability:

- ♦ Scalable across colleges & universities with minimal adaptation.
- ♦ Can evolve into mobile apps, AI analytics, biometric ID integration
- ♦ Aligns with Govt. digital education push (e.g., NEP, Samarth Portal)
- ♦ Ensures data-driven decisions for institutional efficiency

5 Potential Risk and Mitigation Strategies:

 **Risk:** Staff resistance to new system.

 **Solution:** Familiar UI + minimal training + phased rollout.

 **Risk:** Student's Data Privacy & Security

 **Solution:** Role-based access, JWT auth, database encryption, daily backups.

 **Risk:** Integration with Legacy Records

 **Solution:** Automate approvals through AI, workflows and notifications

 **Risk:** Integration complexity with existing systems

 **Solution:** Develop APIs and modular architecture for easy Integration

6 Business Model Ideas

- **Institution Licensing** - Annual/per-student fee for colleges
- **Advertisement & Sponsorship** - Partnerships with ed-tech, job portals
- **Data Insights Services** - Provide anonymized analytics for academic planning



Impact on the Stakeholders:

- **Students:** No queues, online receipts, digital access to all records.
- **Staff:** Elimination of duplicate entries, fewer errors, reduced workload.
- **Administrators:** Real-time analytics for data-driven decisions.
- **Institutions:** Smooth operations, low costs, improved transparency.



Impact on the Target Audience:

- **Efficiency:** Save hours of manual work → faster admissions, fee collection.
- **Accuracy:** Minimize human error → reduce record mismatches.
- **Transparency:** Instant dashboards → higher accountability.
- **Social Impact:** Affordable solution for public colleges → bridging digital divide.

Aspect	Our ERP (TLE Smashers)	TCS iON	SAP Student Lifecycle	Oracle PeopleSoft	Small Vendor Portals
Cost	Low (cloud-native)	Very High (license + infra)	Very High	Very High	Low
Ease of Use	High (simple UI/UX)	Medium	Low (complex UI)	Low	Medium
Cloud Deployment	Yes (Vercel/Heroku)	Partial	Yes	Yes	Rarely
Implementation Time	Quick (modular setup)	Months	Months/Years	Months	Weeks
Modular Adoption	Yes (admissions, fees, hostel, exams separately adoptable)	No	No	No	Limited
Real-time Dashboards	Yes (integrated)	Yes	Yes	Yes	Rarely
Customization	High (open & flexible)	Low (vendor dependent)	Medium	Medium	Low
Support & Training	Minimal training (easy adoption)	Requires vendor training	Requires certified consultants	Requires consultants	Limited support
Suitability for Public Colleges	High (affordable, scalable)	Low	Low	Low	Medium
Data Security	High (role-based access, backups)	High	High	High	Varies
Unique Value	Low-cost, cloud-native, real-time, modular ERP	Established ERP	Enterprise-level	Enterprise-level	Localized but outdated

Fig: Competitive Analysis

9

Research Papers and Articles:

- AI in Student Monitoring:** Uses AI to automate attendance and track activities, improving engagement. [Explore the paper](#)
- Digital Portfolios for Assessment:** Helps students present talents for jobs and evaluation. [Access the research](#)
- Centralized Assignment Platforms:** Links assignment platforms to student success in higher education. [Read the analysis](#)
- Student Tracking Systems:** Provides insights to boost success and optimize resources. [View the report](#)

10

Government Initiatives:

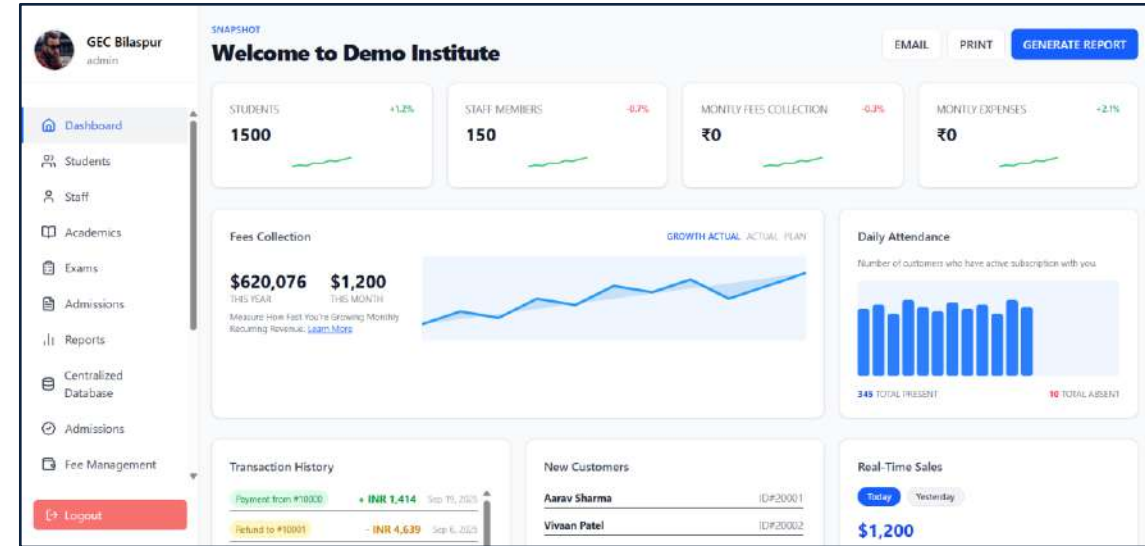
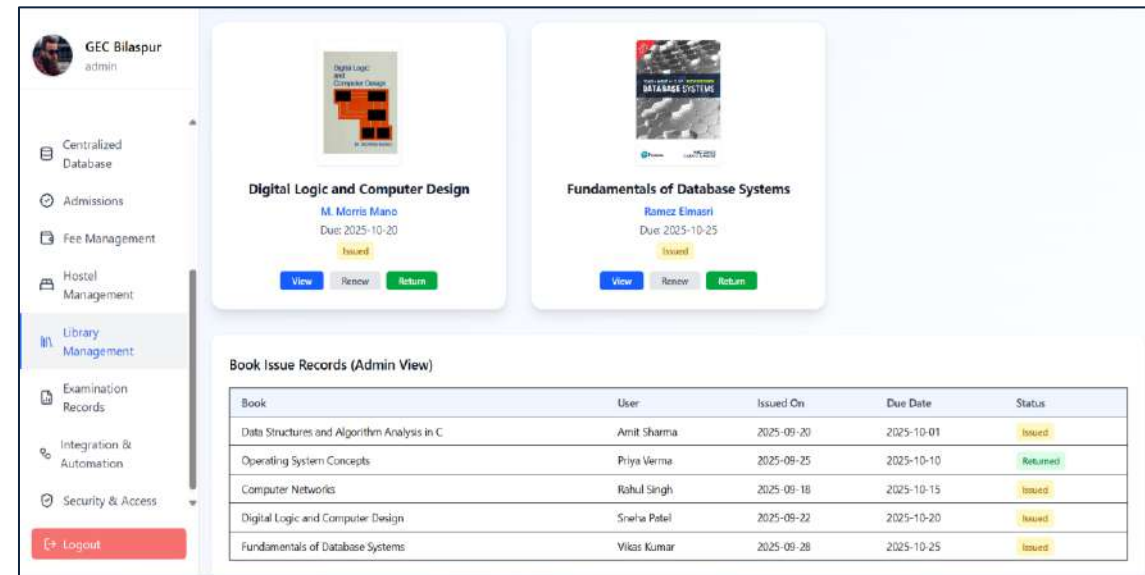
- 'Samarth' Portal (UP):** Digitizes processes in universities for better transparency and efficiency. [Read more](#)



Live Demo Link -



YT Video Link -

Digital Logic and Computer Design

M. Morris Mano
Due: 2025-10-20
Issued

Fundamentals of Database Systems

Ramrez Elmasri
Due: 2025-10-25
Issued

Book Issue Records (Admin View)

Book	User	Issued On	Due Date	Status
Data Structures and Algorithm Analysis in C	Amit Sharma	2025-09-20	2025-10-01	Issued
Operating System Concepts	Priya Verma	2025-09-25	2025-10-10	Returned
Computer Networks	Rohit Singh	2025-09-18	2025-10-15	Issued
Digital Logic and Computer Design	Sneha Patel	2025-09-22	2025-10-20	Issued
Fundamentals of Database Systems	Vikas Kumar	2025-09-28	2025-10-25	Issued